Micro I — Part I: Practice exam 2

Microeconomics I · Master in Economics · Universitat de Barcelona · 2026/27

Name and surname	

Student number										

Rules

- **Part I retake: Friday 18 December 2026**, on the date of the final exam; the Master programme announces the time and the room. 100 points: Part A 40, Part B 30, Part C 30.
- **This practice paper has the same length, format and rules as the Part I exam.** It is not collected or graded; a sample solution follows on the course page.
- **Closed book.** You may use **pen, paper, any handwritten notes** (as many as you like, also photocopied or printed) and a **simple calculator**: basic arithmetic only, not programmable, not graphing, not a phone or any other connected device. You will not need one: every number in this exam works out by hand.
- **No typed or printed text** (typed notes, slides, books). **No formula sheet** is provided. **Phones and all other electronics** are switched off and put away.
- Keep your **photo ID** on the desk; it is checked at the start.
- Any answer that is **not supported by your calculation** receives only partial points or none (this does not apply to the multiple-choice questions).
- If you use additional sheets for calculations, attach them at the end and mark clearly which question each calculation belongs to.
- Do not separate the pages. Write your name and student number on this page only.

Answering

- **Part A** (A1–A12): circle *one* letter in the answer box next to each question; exactly one option is correct; a wrong answer costs nothing. To change an answer, cross out that circle and circle the new letter:



- **Part B** (B1–B7): write the final number in the box and your working in the scratch space below it.
- **Part C**: write your steps in the boxes; each box is marked.
- **Numbers** (Parts B and C): an integer or an exact fraction (such as 7/3, in any equivalent form) is always accepted; you never need a decimal.
- Points are printed next to every question. Notation: page 2.

DO NOT open before start

Notation (rules and answering: page 1)

As in class and in Mas-Colell, Whinston and Green (MWG): bundles $x = (x_1, x_2) \in \mathbb{R}^2_+$, prices $p = (p_1, p_2) \gg 0$, wealth $w > 0$; $x(p, w)$ Walrasian demand, $v(p, w)$ indirect utility, $h(p, u)$ Hicksian demand, $e(p, u)$ expenditure function.

For a price change from p^0 to p^1 with $u^0 = v(p^0, w)$ and $u^1 = v(p^1, w)$: compensating variation $CV = w - e(p^1, u^0)$ and equivalent variation $EV = e(p^0, u^1) - w$, both negative for a loss.

A lottery lists the probabilities of its prizes in the order given. Money is in euros.

Extra space: write which question your working belongs to

Part A · multiple choice, 40 points · circle one letter in the answer box

A1 [3 points]

Bundles $x = (x_1, x_2)$ and $y = (y_1, y_2)$ lie in $\mathbb{R}^2_+$. Ines finds x at least as good as y if and only if $x_1 \geq y_1$ or $x_2 \geq y_2$ (or both). Which statement about Ines's preferences is true?

A They are transitive, but not complete.

B They are neither complete nor transitive.

C They are complete and transitive.

D They are complete, but not transitive.

Answer A1

A B C D

A2 [3 points]

The utility function $u(x_1, x_2) = x_1\,(1 + x_2^2)$ represents a consumer's preferences on $\mathbb{R}^2_+$. Here LNS means locally non-satiated. Which statement about her preferences is true?

A They are monotone, but not strongly monotone.

B They are strongly monotone, hence monotone and LNS.

C They are LNS, but not monotone.

D They are not LNS.

Answer A2

A B C D

A3 [4 points]

Which of these utility functions on $\mathbb{R}^2_+$ is quasiconcave but not concave?

A $u(x_1, x_2) = x_1^{1/3} x_2^{1/2}$

B $u(x_1, x_2) = x_1^2 + x_2^3$

C $u(x_1, x_2) = \sqrt{x_1 + 2x_2}$

D $u(x_1, x_2) = x_1^{2/3} x_2^{2/3}$

Answer A3

A B C D

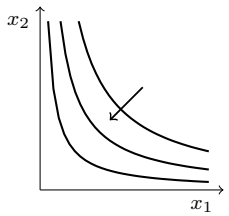
A4 [3 points]

Each panel shows three indifference curves of a preference on $\mathbb{R}^2_+$; the arrow points towards better bundles. Which panel shows preferences that are convex, but not monotone? Circle that panel's letter in the answer box.

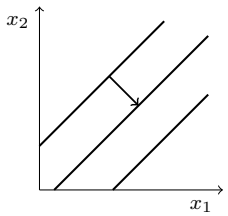
Answer A4

A B C D

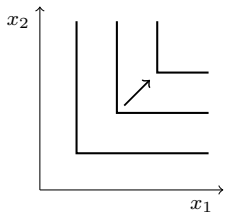
A



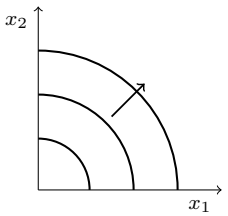
B



C



D



A5 [3 points]

Noa's preferences are rational, continuous, locally non-satiated and convex (not necessarily strictly convex). At prices $p = (5, 2)$ and wealth $w = 40$, both bundles $(2, 15)$ and $(6, 5)$ are in her Walrasian demand $x(p, w)$. Which bundle must also be in $x(p, w)$?

A $(4,\ 9)$

B $(4,\ 10)$

C $(5,\ 10)$

D $(8,\ 0)$

Answer A5

A B C D

A6 [4 points]

Prices are $p = (p_1, p_2) \gg 0$ and wealth is $w > 0$. Which function could be the indirect utility function $v(p, w)$ of a consumer with locally non-satiated preferences?

A $v(p, w) = \dfrac{w^2}{p_1 p_2 + p_2^2}$

B $v(p, w) = \dfrac{p_1 + p_2}{w}$

C $v(p, w) = \dfrac{w^3}{p_1 p_2 + p_2^2}$

D $v(p, w) = \dfrac{w\, p_1}{p_2^2}$

Answer A6

A B C D

Scratch for A6 (not marked)

A7 [4 points]

Two goods. A consumer's preferences are rational, continuous, locally non-satiated and strictly convex; her Walrasian demand $x(p, w)$ and Hicksian demand $h(p, u)$ are differentiable. At prices p and wealth w she buys the bundle $x(p, w) = (x_1, x_2) = (4, 6)$. There, $\partial x_1/\partial p_1 = -3$, and at $u = v(p, w)$ her compensated slope is $\partial h_1/\partial p_1 = -1$. What is $\partial x_1/\partial w$ at (p, w)?

A $-\frac{1}{2}$

B 2

C $\frac{1}{2}$

D $\frac{1}{3}$

Scratch for A7 (not marked)

Answer A7

A B C D

A8 [4 points]

Two goods. A consumer's preferences on $\mathbb{R}^2_+$ are rational, continuous, locally non-satiated and strictly convex; her Walrasian demand $x(p, w)$ and Hicksian demand $h(p, u)$ are differentiable, and she always buys positive amounts of both goods. Which statement must be true at every price vector p, wealth w and utility level u?

A $\partial x_1/\partial p_2 \geq 0$: the goods are gross substitutes.

B $\partial x_1/\partial p_2 = \partial x_2/\partial p_1$.

C $\partial h_1/\partial p_1 > 0$ wherever good 1 is a Giffen good.

D $\partial h_1/\partial p_2 \geq 0$: the goods are net substitutes.

Answer A8

A B C D

A9 [3 points]

Prizes are 80, 35 and 0 euros; a lottery lists their probabilities in this order. With probability $\frac{1}{3}$ Theo plays $L_1 = (\frac{1}{3},\ 0,\ \frac{2}{3})$; otherwise he plays $L_2 = (\frac{2}{3},\ \frac{1}{3},\ 0)$. What is the reduced lottery of this compound lottery?

A $(\frac{2}{3},\ \frac{1}{3},\ 0)$

B $(\frac{1}{2},\ \frac{1}{6},\ \frac{1}{3})$

C $(\frac{5}{9},\ \frac{2}{9},\ \frac{2}{9})$

D $(\frac{4}{9},\ \frac{1}{9},\ \frac{4}{9})$

Scratch for A9 (not marked)

Answer A9

A B C D

A10 [3 points]

Prizes are 300, 120 and 0 euros; a lottery lists their probabilities in this order. Sam strictly prefers $L_1 = (0,\ 1,\ 0)$, 120 euros for sure, to $L_1' = (\frac{2}{9},\ \frac{2}{3},\ \frac{1}{9})$. He also strictly prefers $L_2' = (\frac{2}{9},\ 0,\ \frac{7}{9})$ to $L_2 = (0,\ \frac{1}{3},\ \frac{2}{3})$. Which statement is true?

A His choices fit expected utility with a concave Bernoulli utility.

B His choices violate the transitivity axiom.

C His choices violate the independence axiom.

D His choices fit expected utility with a convex Bernoulli utility.

Scratch for A10 (not marked)

Answer A10

A B C D

A11 [3 points]

Uma is an expected utility maximizer with Bernoulli utility $u(x)$, where $u(x) > 0$ for every prize x. Ben uses a Bernoulli utility $\tilde{u}$ built from u. With which $\tilde{u}$ does Ben rank every pair of lotteries exactly as Uma does, whatever u is?

A $\tilde{u}(x) = -1/u(x)$

B $\tilde{u}(x) = \frac{1}{2}\,u(x) + 3$

C $\tilde{u}(x) = 2^{u(x)}$

D $\tilde{u}(x) = 3 - \frac{1}{2}\,u(x)$

Answer A11

A B C D

A12 [3 points]

Kofi is an expected utility maximizer. His Bernoulli utility over money $x \geq 0$ in euros is $u(x) = x/(x+10)$. A lottery pays 90 euros with probability $\frac{2}{3}$ and 0 euros with probability $\frac{1}{3}$. What is its risk premium RP (expected value minus certainty equivalent), in euros?

A 45

B 60

C 54

D 0

Scratch for A12 (not marked)

Answer A12

A B C D

Part B · numerical answers, 30 points · final number in the box, working below

B1 [4 points]

Preferences on $\mathbb{R}^2_+$ are represented by $u(x_1, x_2) = x_1\sqrt{x_2} + 2x_2$. The marginal rate of substitution is $\mathrm{MRS}_{12}(x) = \dfrac{\partial u/\partial x_1}{\partial u/\partial x_2}$. What is MRS_{12} at the bundle $x = (3, 9)$?

Answer B1: $\mathrm{MRS}_{12}(3, 9) =$

Your working (it must support the answer in the box)

B2 [4 points]

Preferences on $\mathbb{R}^2_+$ are represented by $u(x_1, x_2) = x_1^2 + 3x_1x_2$. Find the number x_2 such that the bundle $(2, x_2)$ is indifferent to the bundle $(4, 2)$.

Answer B2: $x_2 =$

Your working (it must support the answer in the box)

B3 [5 points]

A consumer has utility $u(x_1, x_2) = x_1 + 3x_2^{1/3}$ on $\mathbb{R}^2_+$ (useful: $8^{1/3} = 2$). Prices are $p = (p_1, p_2) = (8, 2)$ and her wealth is $w = 9$. How many units of good 2 does she buy?

Answer B3: $x_2 =$

Your working (it must support the answer in the box)

B4 [4 points]

A consumer's indirect utility function is $v(p, w) = 2 \ln w - \ln p_1 - \ln(p_1 + p_2)$. Use Roy's identity: what is her demand for good 1 at prices $p = (6, 3)$ and wealth $w = 18$?

Answer B4: $x_1 =$

Your working (it must support the answer in the box)

B5 [5 points]

A consumer's expenditure function is $e(p,u) = u\sqrt{p_1^2 + 3p_1p_2}$. What is her Hicksian demand for good 1, $h_1(p,u)$, at prices $p = (1,8)$ and utility level $u = 5$?

Answer B5: $h_1 =$

Your working (it must support the answer in the box)

B6 [4 points]

Sara is an expected utility maximizer with Bernoulli utility $u(x) = x^{2/3}$ over money $x \geq 0$ in euros (useful: $8^{1/3} = 2$, $27^{1/3} = 3$, $64^{1/3} = 4$). A lottery pays 64 euros with probability $\frac{1}{5}$ and 1 euro with probability $\frac{4}{5}$. What is its certainty equivalent, in euros?

Answer B6: certainty equivalent $=$

Your working (it must support the answer in the box)

B7 [4 points]

Pia is an expected utility maximizer with Bernoulli utility $u(x) = (x+8)^{1/3}$ over money $x \geq 0$ in euros (useful: $27^{1/3} = 3$, $64^{1/3} = 4$, $125^{1/3} = 5$). A lottery pays 117 euros or 19 euros, each with probability $\frac{1}{2}$. What is its risk premium RP (expected value minus certainty equivalent), in euros?

Answer B7: $RP =$

Your working (it must support the answer in the box)

Problem C1 [16 points]

A consumer has the utility function $u(x_1, x_2) = (x_1 + 3)\, x_2$ on $\mathbb{R}^2_+$. Write your steps in the boxes and the results in the answer boxes.

(a) [5 points] Find the Walrasian demand $x(p, w) = (x_1(p, w), x_2(p, w))$ for $w > 3p_1$.

steps for (a)

$x_1(p, w) =$ $x_2(p, w) =$

(b) [3 points] Find the indirect utility function $v(p, w)$ for $w > 3p_1$.

steps for (b)

$v(p, w) =$

(c) [4 points] Find the expenditure function $e(p, u)$ for $u > 9p_1/p_2$, and the Hicksian demands $h_1(p, u)$ and $h_2(p, u)$.

steps for (c)

$e(p, u) =$

$h_1(p, u) =$ $h_2(p, u) =$

(d) [4 points] Let $p_1 = 1$ and $w = 13$, and let the price of good 2 rise from $p_2^0 = 1$ to $p_2^1 = 4$. Compute her compensating variation CV and her equivalent variation EV (as defined on page 2) of this price change.

steps for (d)

$CV =$ $EV =$

Problem C2 [14 points]

A consumer's preferences on $\mathbb{R}^2_+$ are represented by $u(x_1, x_2) = (2x_1 + x_2)^2 + 2x_1 + x_2$. Write your answers in the boxes.

(a) [3 points] Are the preferences represented by u strongly monotone? Circle Yes or No. Prove your answer, or give two bundles that show it.

Circle: Yes No

reason or counterexample for (a)

(b) [3 points] Are these preferences convex? Circle Yes or No and explain.

Circle: Yes No

reason for (b)

(c) [4 points] Is the function u concave? Circle Yes or No. Prove your answer, or give two bundles and their midpoint that show it.

Circle: Yes No

proof, or two bundles and their midpoint, for (c)

(d) [4 points] Give a concave utility function $\tilde{u}$ that represents the same preferences. Explain why it represents them and why it is concave.

$\tilde{u}(x_1, x_2) =$

why it represents the same preferences, and why it is concave

Extra space: write which question your working belongs to. If you use additional sheets for calculations, attach them at the end and mark clearly which question each calculation belongs to.

Extra space: write which question your working belongs to. If you use additional sheets for calculations, attach them at the end and mark clearly which question each calculation belongs to.